SEC 1 · SATELLITE TYPES & ORBITS

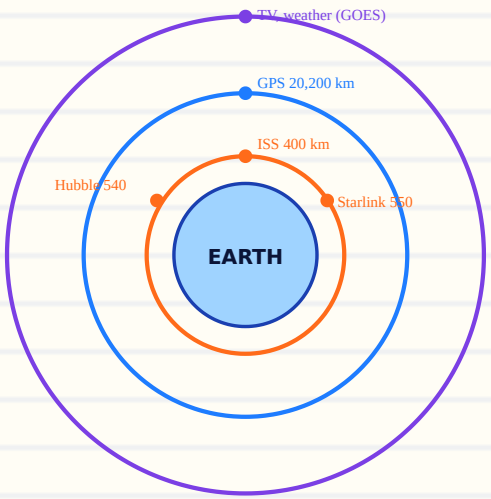
Your spread, upgraded

Your right page was already the correct shape. This is the same content with a drawn left page, harder cues, grouped lists, and the summary box. Copy what helps.

LEFT PAGE · DRAW

SEC 1 — SATELLITE TYPES & ORBITS

date · pg __



GEO = 35,786 km
1 orbit = 1 day → hangs over one spot
(geoSTATIONARY if also over the equator)

MEO = 2,000 – 35,786 km
GPS: 12-hour orbit

LEO = 180 – 2,000 km
~90-minute orbit, 7.7 km/s
drag + atomic oxygen live here

rule: higher = slower = longer day
LEO 90 min → MEO 12 h → GEO 24 h

not to scale! GEO is ~6 Earth-radii out →

⭐ 3 TAKEAWAYS

1. Higher orbit = slower = longer trip around (90 min → 12 h → 24 h)
2. GEO is special only because its day matches Earth's day
3. Group the 8 satellite types by JOB: look out / look down / relay / live

RIGHT PAGE · CORNELL

CUES = questions | **NOTES = keywords**

Which satellite types **look out**, which **look down**, which **relay**?

Where does the ISS live? GPS? TV satellites?

Why is GEO exactly 35,786 km?

Geosynchronous vs geostationary?

Higher orbit → faster or slower?

? what orbit do spy sats use, and why

LOOK OUT — astronomical (Hubble, JWST)
LOOK DOWN — Earth obs (Landsat), weather (GOES), recon (spy)
RELAY — comms (Starlink, DirecTV), navigation (GPS)
LIVE — biosatellites (Laika!), space stations (ISS)

LEO 180–2000 km · ISS 400 · ~90 min
MEO 2000–35,786 · GPS 20,200 · 12 h
GEO 35,786 · 1 orbit = 1 Earth day → stays put
geoSYNC = 1-day orbit, any tilt. geoSTATIONARY = 1-day + over equator
higher = slower (bigger circle, weaker gravity)

job table:

type	job (5 words)	example
navigation	tells you where you are	GPS
comms	bounces signals across Earth	Starlink
weather	watches clouds from GEO	GOES
…	…	…

🧠 SUMMARY — video closed, from memory (5 lines)

1. _____
2. _____
3. _____
4. _____
5. _____

FEEDBACK · YOUR PAGE

What was right, what to sharpen

Honest read of the photo you sent. The structure was correct on the first try, which is the hard part.

✔ Correct

- **Cues are questions.** "What are the types of satellites?" not "Satellite types." That's the thing most people never do.
- **Keywords, not sentences.** Nothing to fix there.
- **All three altitude ranges are right.** LEO 180–2,000 km, MEO 2,000–35,786 km, GEO 35,786 km.
- **You started a third cue mid-lecture** ("job of each satellite"). That's exactly how the column is supposed to grow.

▲ Sharpen

- **Geosynchronous ≠ geostationary.** Geosynchronous = the orbit takes one day, at any tilt. Geostationary = one day AND over the equator, so it hangs over one spot. TV and weather sats are geostationary. Write both words. Classic quiz trap.
- **Add the one "why."** GEO is 35,786 km because at that height one orbit takes exactly one Earth rotation (23 h 56 min). Lower = faster, higher = slower. Every other orbit number you'll just memorize; this one you can explain.
- **Cues are too easy.** "What are the orbit types?" you can answer with half a glance. Use cues you might fail: "Where does GPS live?", "Why 35,786?", "Higher = faster or slower?"
- **Eight-item list → four groups.** Flat lists of 8 don't retrieve. "Look out / look down / relay / live" does.
- **Job table = type · job · one real example.** The example is what sticks.

🖍 Left page was blank

This lecture is a picture: Earth, three rings, residents on each ring. Page 1 shows it. Draw yours badly first, then fix it. Don't copy page 1 neatly — the drawing is the memory, the copying isn't.

🧠 Missing: header + summary

- Top of page: section / lecture / date, so the index can point here.
- Bottom: five lines from memory, video closed. Can't fill five? That's your first gap. Good — now you know where it is.

📒 For the Question bank (back pages)

- Why is GEO exactly 35,786 km?
- Geosynchronous vs geostationary: which one hangs over one spot?
- Name the orbit and altitude of: ISS, GPS, a TV satellite.
- Group the 8 satellite types by job.

⚡ Next step: finish the lecture, write the 5-line summary, send me a photo. I'll check it and write the next cards.

SEC 1 · VOCABULARY

Satellite types & orbits, in plain English

Every word used on the previous pages, including the ones I introduced in the notes column. Read once; on the +1 day review, explain three of them out loud without looking.

Satellite — anything that orbits something bigger. The Moon is Earth's natural satellite; everything we launch is an artificial one. *A ball on a string, swung so fast the string is gravity.*

Altitude — height above the ground, not from Earth's center. GEO is 35,786 km *altitude*; its distance from the center is that plus Earth's 6,371 km radius.

Period — how long one full trip around takes. LEO ~90 min, GPS 12 h, GEO 24 h. Higher = longer. *Bigger circle, slower runner.*

Geostationary — geosynchronous AND directly over the equator with no tilt, so it appears frozen in one spot. TV dishes point at these and never move. *Sync = same time every day. Stationary = never leaves.*

Sidereal day — one true rotation of Earth relative to the stars: 23 h 56 min, not 24 h. That's the real period a GEO satellite matches. *The 4-minute difference is Earth also moving around the Sun.*

Earth observation (EO) — cameras and radar pointed down at Earth for mapping, farming, disasters (Landsat). *Looks down.*

Reconnaissance satellite — a military spy satellite. Same idea as Earth observation, different customer and classified detail.

Communication satellite — a radio relay in the sky: receives a signal from one place on Earth and sends it back down somewhere else (Starlink, DirecTV). *A mirror for radio.*

Space station — a satellite people live in (ISS, China's Tiangong).

GPS — Global Positioning System, the US navigation constellation at ~20,200 km in MEO.

Starlink — SpaceX's internet constellation: thousands of small comms satellites in LEO at ~550 km.

Eclipse (orbital) — when Earth blocks the Sun from the satellite. LEO sats get ~15 a day; batteries have to cover them.

Orbit — the closed path a satellite follows around a body. It's falling toward Earth and missing, forever. *Throw a ball sideways fast enough and the ground curves away as fast as it falls.*

LEO / MEO / GEO — Low, Medium, Geosynchronous Earth Orbit: the three height bands. LEO 180–2,000 km, MEO 2,000–35,786 km, GEO 35,786 km. *Downstairs, middle floor, penthouse.*

Geosynchronous — an orbit whose period is exactly one day, at any tilt. The satellite returns to the same spot in the sky once a day but may drift north-south in between.

Equator — the imaginary line around Earth's middle. A geostationary satellite has to sit above it; a satellite above Texas can't hover there because Earth's spin would carry Texas away underneath.

Astronomical satellite — a telescope in space pointed away from Earth (Hubble, JWST). *Looks out.*

Weather satellite — looks down at clouds and storms, usually from GEO so it watches the same region nonstop (GOES).

Navigation satellite — broadcasts a precise time signal; your phone hears four of them and works out where it is (GPS, Galileo). *Tells you where you are.*

Biosatellite — carries living things (plants, animals, cells) to study how they cope with space. Sputnik 2 carried Laika the dog in 1957.

ISS — the International Space Station, in LEO at ~400 km, orbiting every ~90 minutes.

Hubble / JWST — two famous space telescopes. Hubble orbits Earth at ~540 km; JWST sits 1.5 million km away at a balance point called L2, which is why it isn't in any of the three bands.

Landsat / GOES — Landsat = US Earth-observation series since 1972 (LEO). GOES = US weather satellites (GEO).

Atomic oxygen / drag — two LEO-only problems: single oxygen atoms that chew plastics, and thin air that slowly slows the satellite down. Both vanish above ~1,000 km. *Covered in the Space Environment section.*